Modelling Reward-Based Decision-Making under Stress and Uncertainty

Yatin Azad

Table of contents

01 Research Problem

02 Methodology

03 Key Results

About me

Personal profile

- Sr. Project Associate at Decision Lab, IIT Kanpur.
- Working on maladaptive decision-making in the presence of stressors and uncertain environments.
- B.Tech in Biological Sciences and Bioengineering from IITK (2017-2021).
- Leader of brain and cognitive society at IITK
- Enjoy board games, badminton and cooking.

Keystone Projects

Decision-Making

Understanding value-based decision-making under stress and uncertainty

Virology

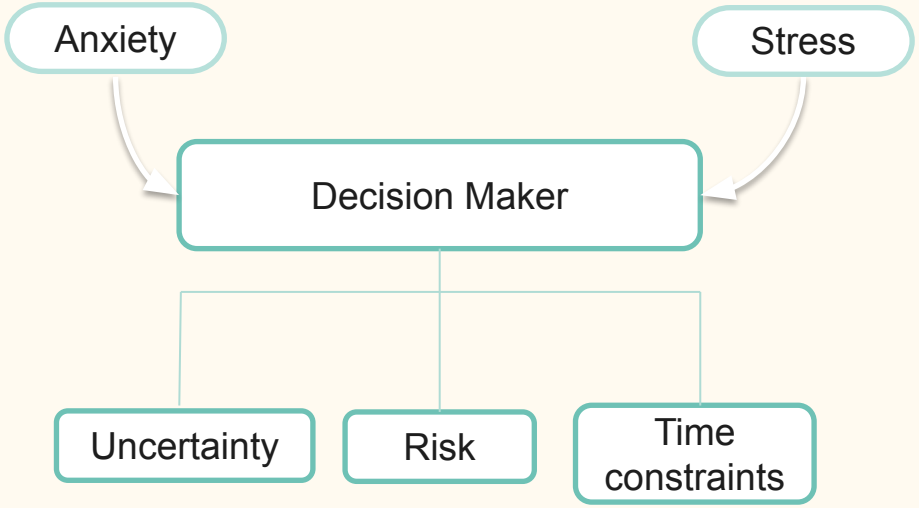
Single-particle tracking of SARS Cov-2 virions inside mammalian cells

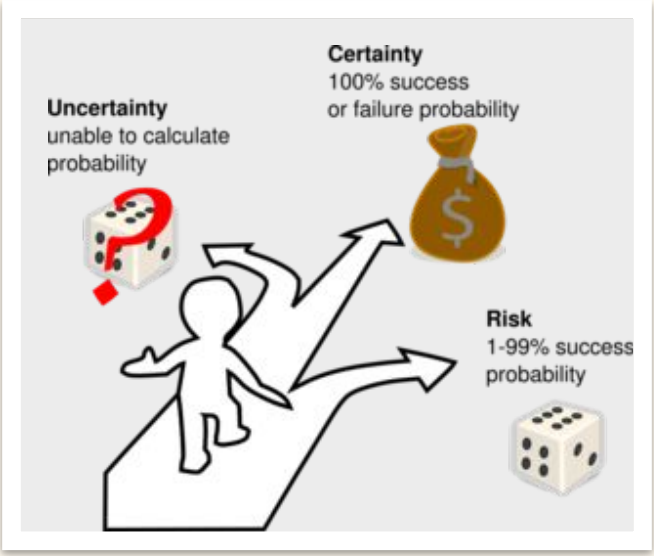
Treadwill

Cognitive disorders

Using unguided computerised cognitive behavioural therapy to treat mood disorders

Investigating 3-way interaction between stress, anxiety, uncertainty.



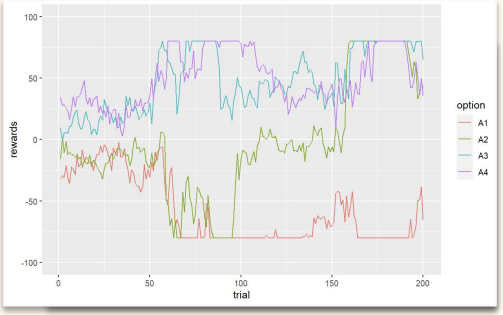


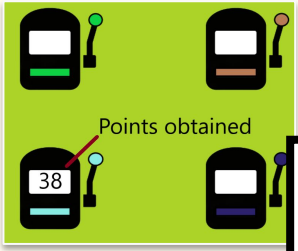
Task Design

- Adapted a version of the restless bandit task used by Speekenbrink & Konstantinidis, 2015

- 4 experimental conditions-:

  a) Stable volatility with loss
  b) Stable volatility without loss
  c) Variable volatility with loss
  d) Variable volatility without loss

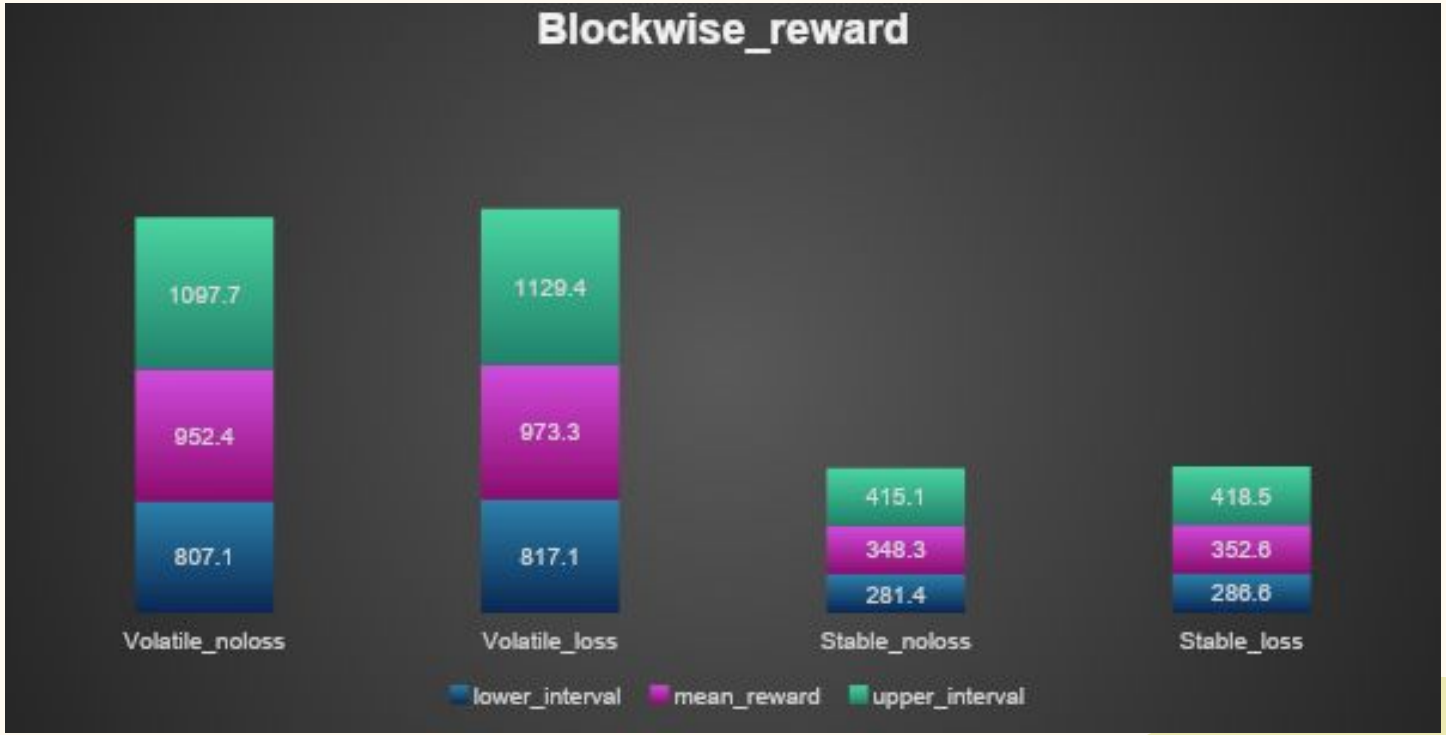




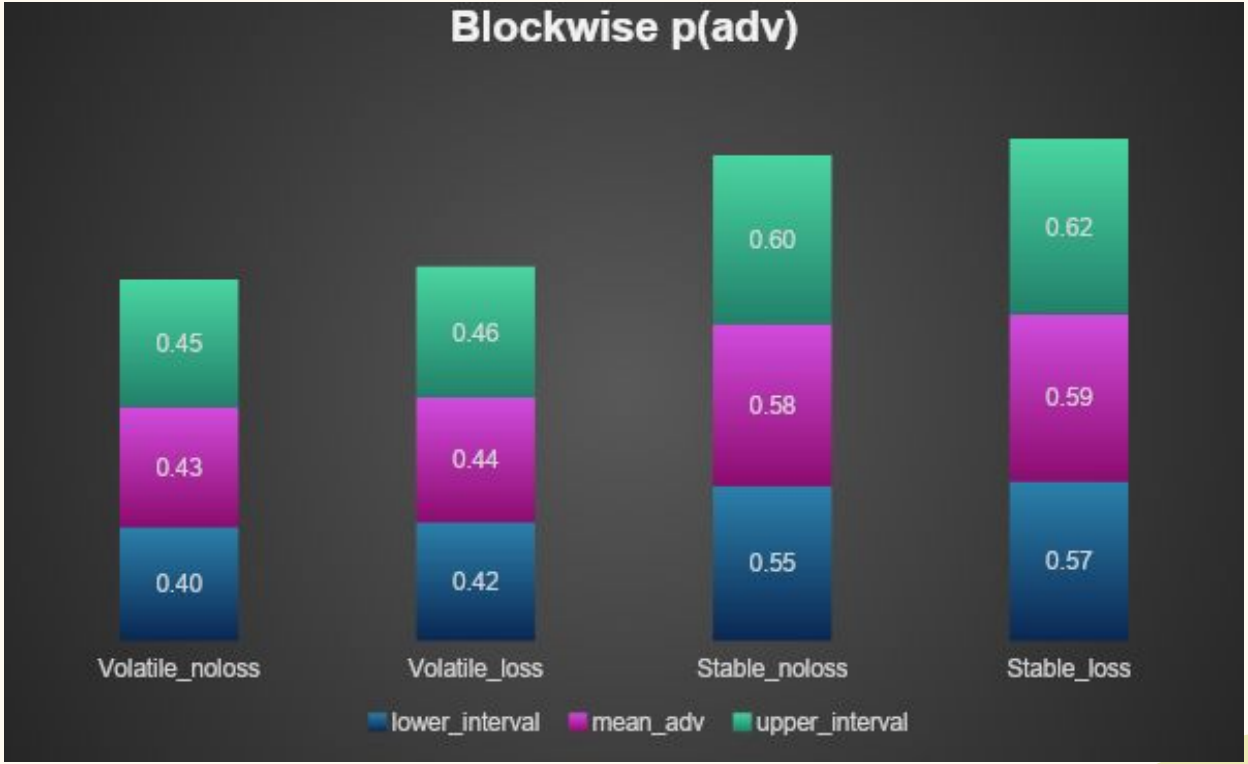
$$r_j(t) = \mu_j(t) + \epsilon_j(t) \qquad \epsilon_j(t) \sim N(0, \sigma_\epsilon)$$

$$\mu_j(t) = \lambda\mu_j(t-1) + \zeta_j(t) \qquad \zeta_j(t) \sim N(0, \sigma_\zeta(t))$$

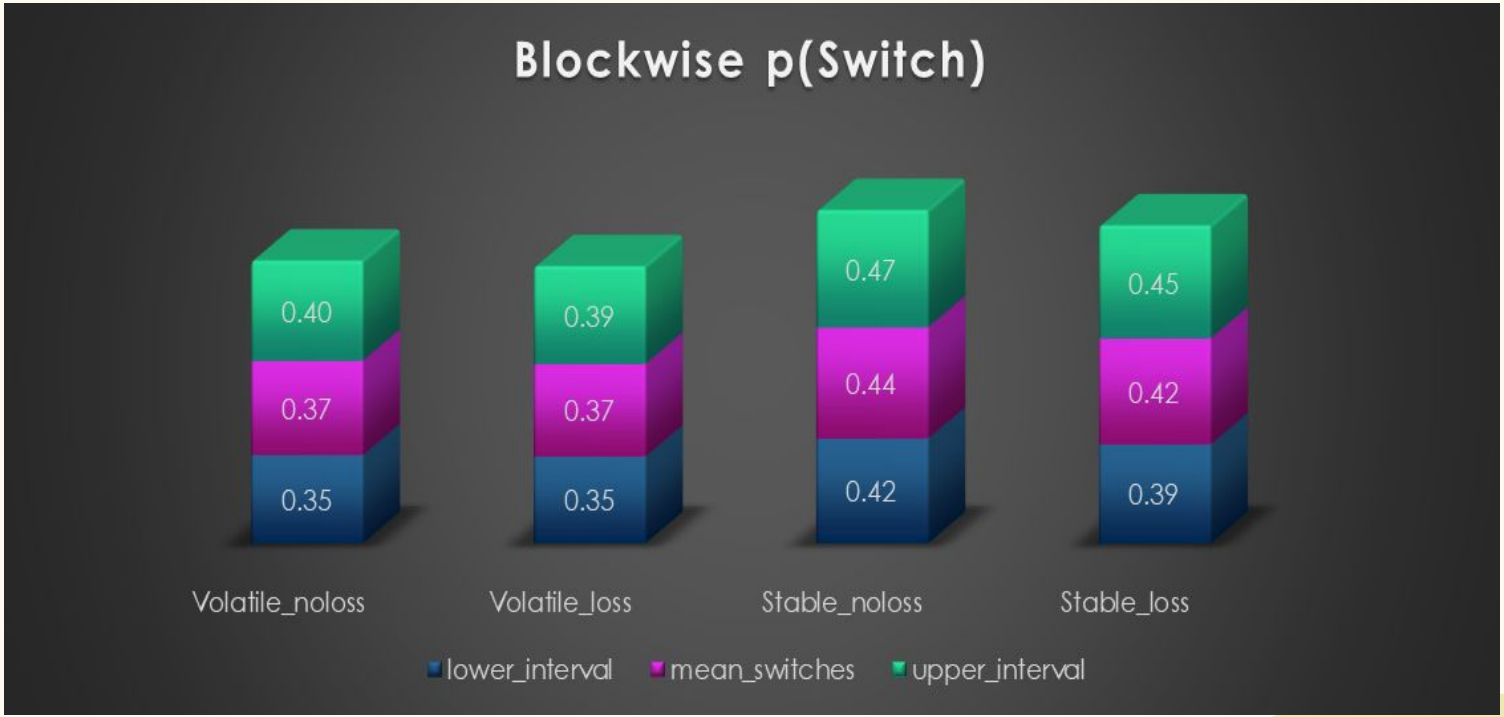
Reward Statistics across experimental conditions



Volatile Environments Reduce Likelihood of Optimal Choices



Volatile Environments Impact Consistency of Decisions

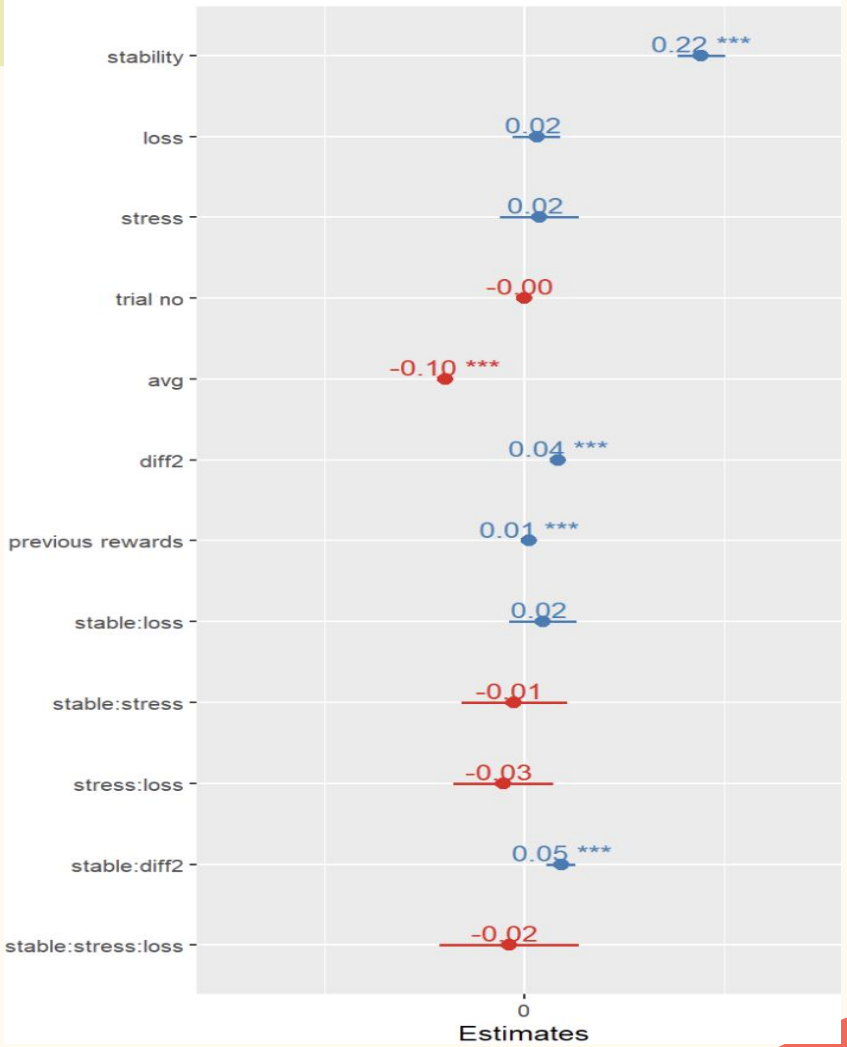


Best Models for Predicting Reward Seeking Behaviour

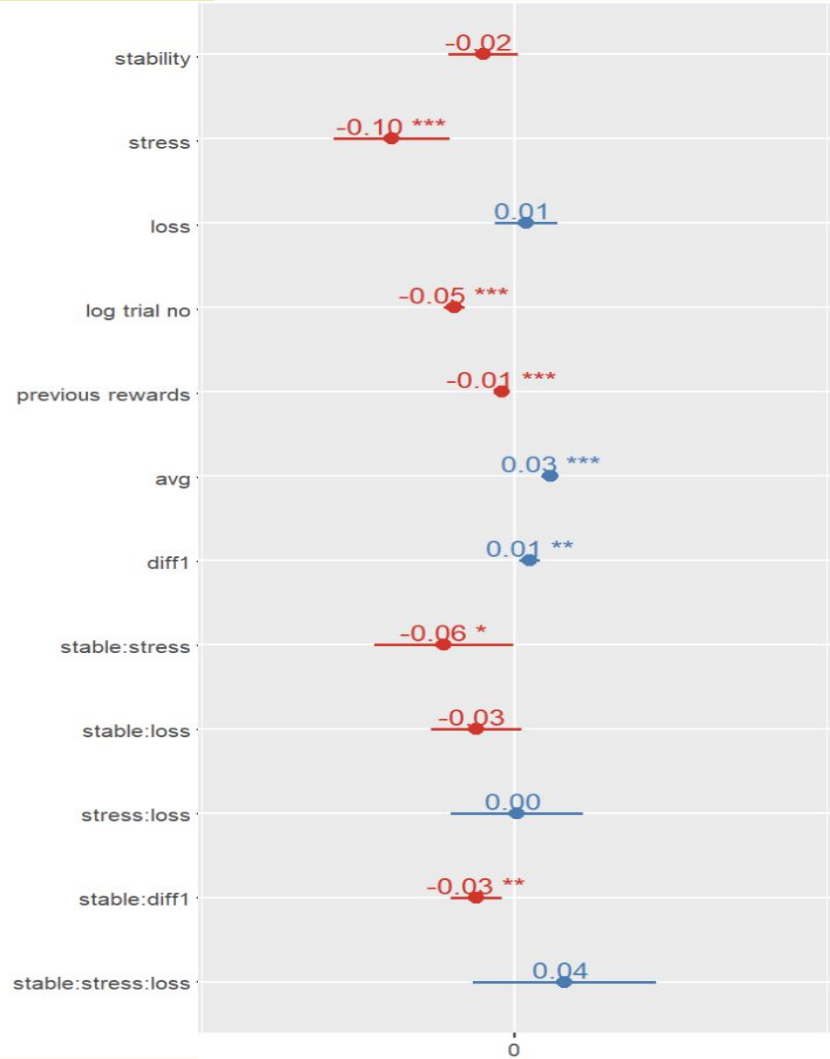
advantageous ~ loss * volatility * stress + reward + trial + ID

switch ~ loss* volatility *stress + reward + trial + ID

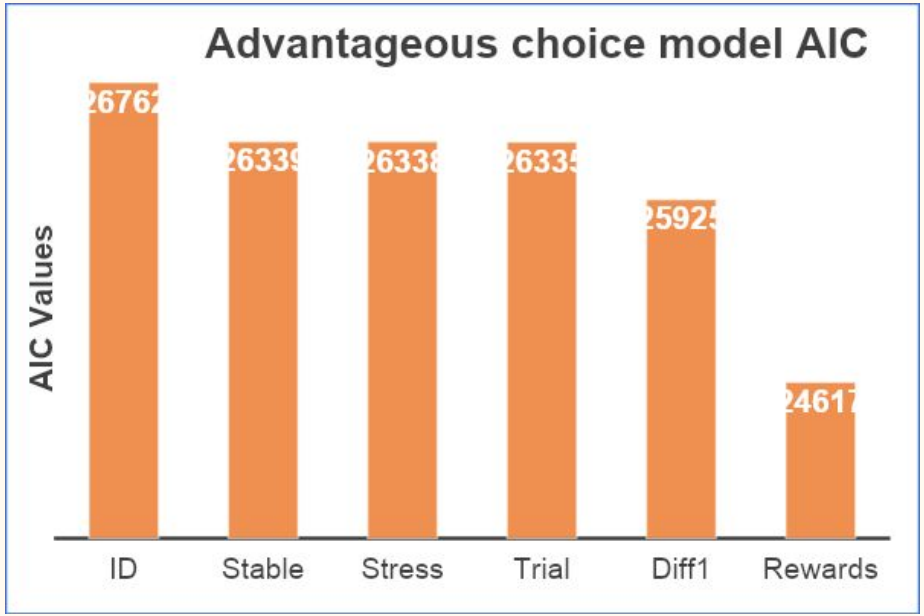
Regression Estimates for Optimal Choice

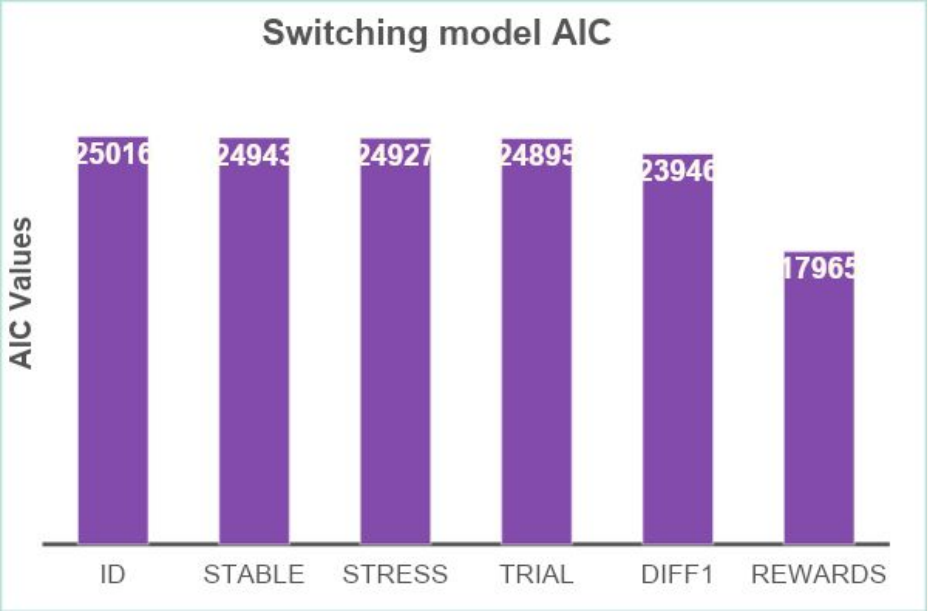


Regression Estimates for Switching Behaviour

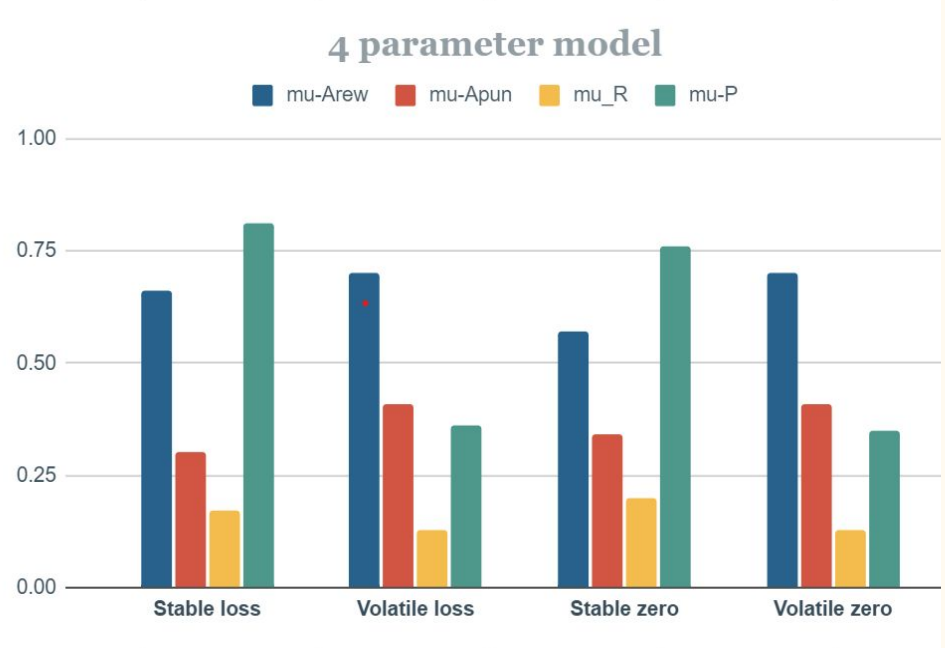


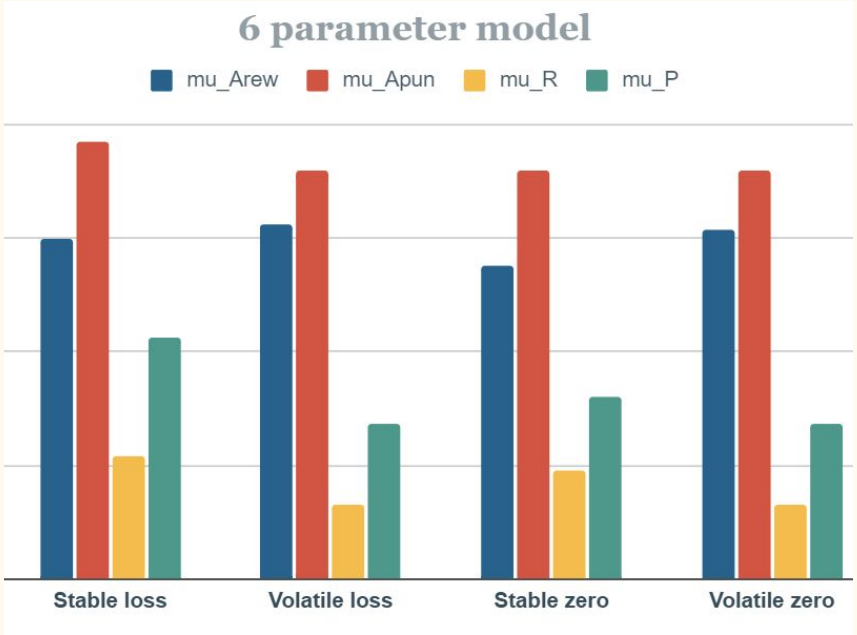
Comparing Across Models



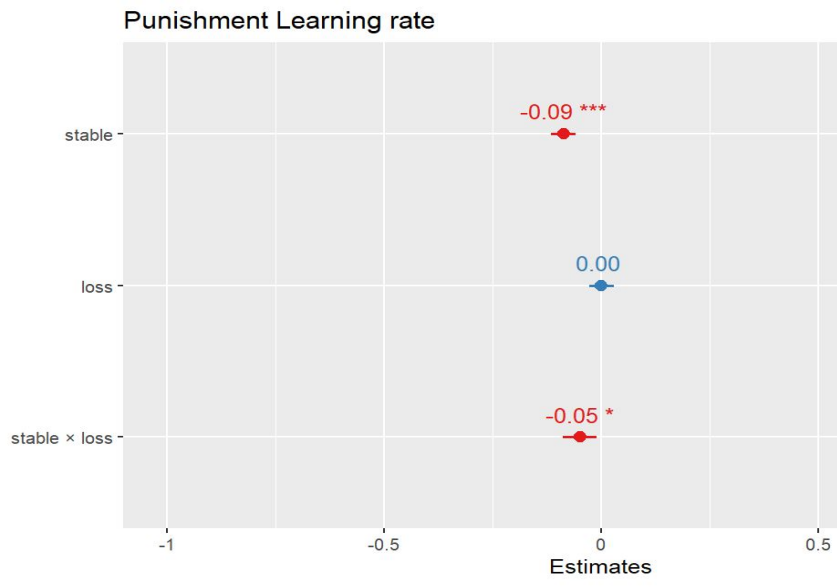


Population-wide Differences in Reward and Punishment Sensitivity

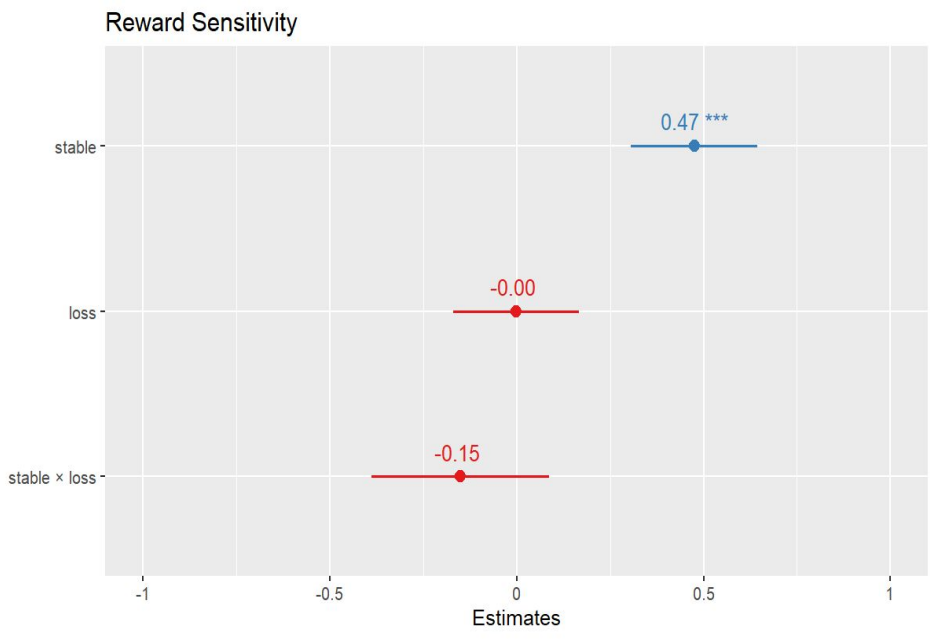


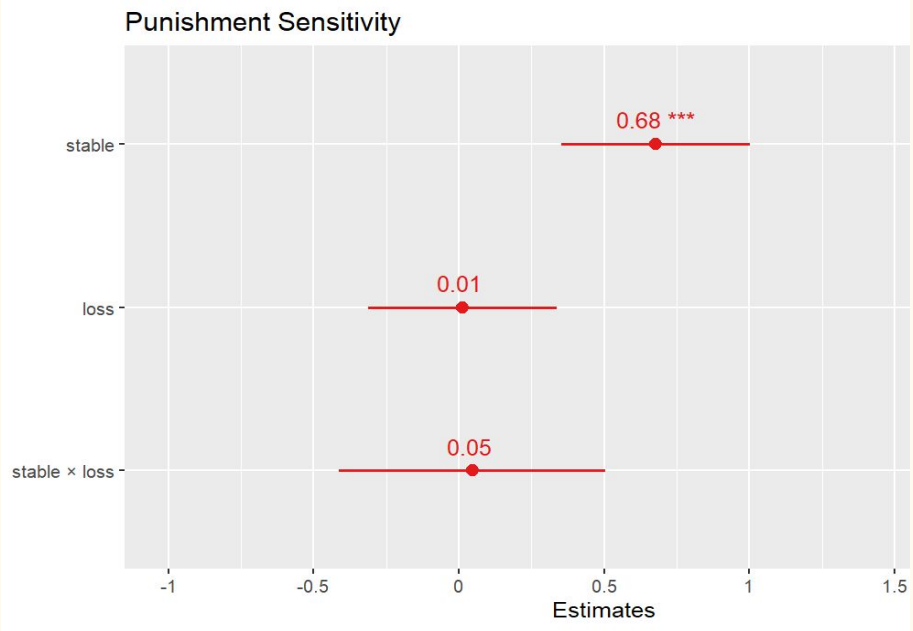


Learning Behaviour Significantly Affected by State

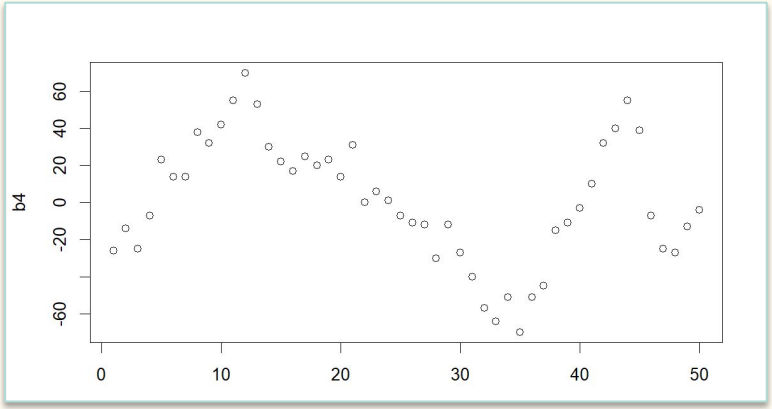


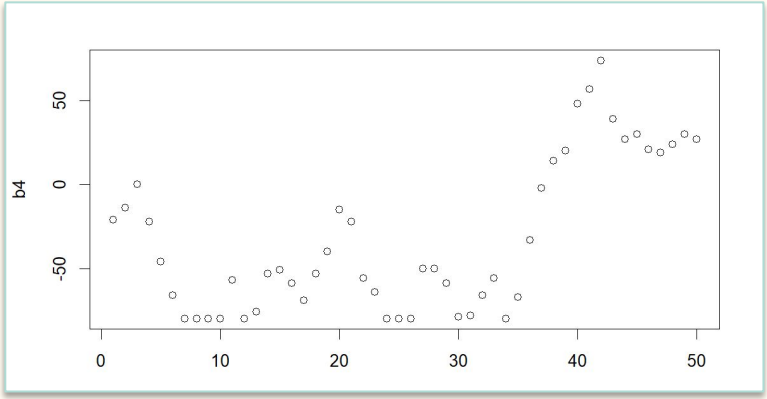
Reward and Punishment Sensitivity Significantly Affected by State

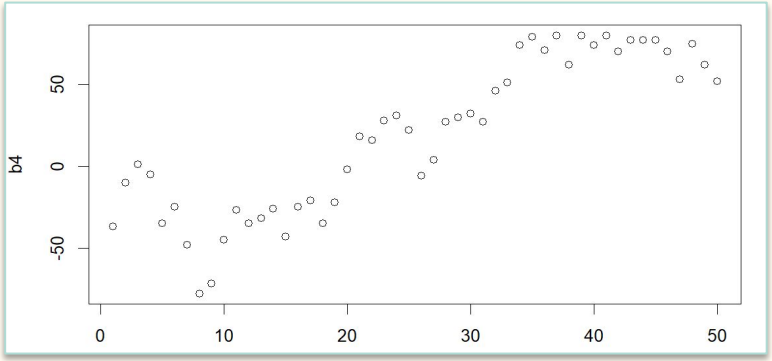


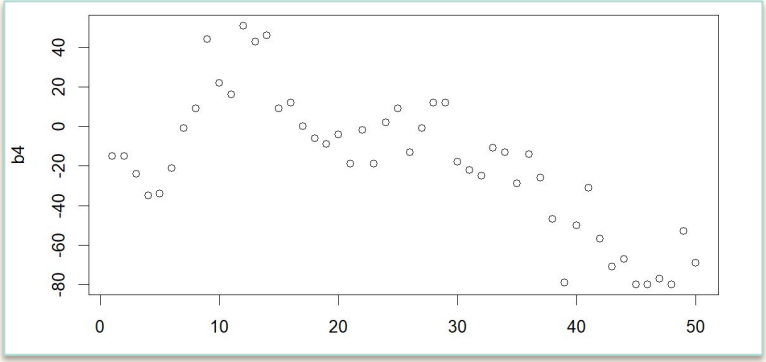


Individual Differences in Reward Seeking Behaviour









Highlights

- ✓ Participants learned over trials. Explored less and performance improved.
- ✗ Performed better in stable blocks.
- ✓ Higher stress correlated with less exploration.
- ✗ Monetary loss at the start led to more optimal choices
- ✓ High positive rewards in preceding trial made them unlikely to switch.

Thanks!